

Surface mass balance inference from the inversion of a glacier evolution model.

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ABSTRACT. The distributed surface mass balance (SMB) of mountain glaciers is a key forcing for projecting glacier evolution and subsequent meltwater availability. Yet, direct in situ observations cover less than 0.2 % of the world’s glacierised area. Here, we present a new SMB inference method, implemented within the open-source Instructed Glacier Model (IGM), that recasts the problem as a fully differentiable transient inversion. We first assimilate observed surface ice speeds at one snapshot, then run a forward simulation between two epochs of known surface elevation under a candidate SMB. Automatic differentiation through the forward run yields the gradient of a geometry-misfit cost function with respect to the SMB, enabling gradient-based optimisation of the latter. We apply the method to three glaciers in the Alps—Argentièrre, Great Aletsch and Rhône—and validate the inferred profiles against in situ stake measurements, outperforming the distributed reconstruction of Kneib and others (2024) on Argentièrre. A sensitivity test shows that the surface-velocity misfit alone bounds the inferred SMB on stake-free glaciers. Our method is generic, computationally efficient and requires no glacier-specific tuning, making it readily transferable to any glacier with accurate multi-temporal DEMs and surface velocities, opening a path toward regional-scale SMB inversion.

INTRODUCTION

Mountain glaciers are sensitive recorders of climate and a critical component of regional water resources, yet projecting their evolution requires their distributed surface mass balance (SMB)—the spatial pattern of accumulation and ablation—which remains poorly constrained at the global scale. Direct, in situ measurements rely on stake networks and snow pits that are labour-intensive, hazardous in steep accumulation areas, and deployed on fewer than 0.2 % of the world’s glacierised area (Cogley and others, 2011). Regional and global climate models provide continuous gridded SMB but reproduce glacier-scale mass change only after calibration (Radić and others, 2014; Marzeion and others, 2012). Most glaciers therefore remain effectively unmonitored, and regional-to-global mass-balance assessments depend on extrapolation from a few benchmark sites or on model reconstructions (Hugonnet and others, 2021; Zemp and others, 2019). As remote-sensing products of glacier surface velocity (Millan and others, 2022), ice thickness (Farinotti and others, 2019; Millan and others, 2022) and elevation change (Hugonnet and others, 2021; Beraud and others, 2022) reach near-global coverage, the question is no longer whether the data exist, but how to combine them to infer physically consistent SMB fields.

Existing approaches

SMB can be derived directly from climate models that simulate surface energy balance and precipitation, but their coarse resolution and simplified treatment of high-mountain microclimate bias precipitation phase, accumulation gradients and melt, so they rarely reproduce observed glacier change without additional bias correction (Marzeion and others, 2012; Rounce and others, 2020). Alternatively, sparse direct measurements are extended to distributed fields by homogenising them against decadal geodetic volume change, as in the long-running Glacier Monitoring Switzerland programme (GLAMOS; ?); this yields a valuable observational reference, but a fundamentally different product from the transient inversion proposed here.

The route most closely related to the present work exploits the mass-conservation equation, which relates the rate of surface-elevation change to the divergence of the ice flux and the SMB (Cuffey and Paterson, 2010). Given ice thickness and surface velocity, the vertically-integrated flux is computed, differentiated spatially, and combined with an observed elevation-change field to yield the SMB. From early flux-gate estimates of emergence velocity in the ablation zone (Berthier and Vincent, 2012; Brun and others, 2018), the approach has been generalised to fully distributed SMB over entire glaciers (Bisset and others, 2020; Miles and others, 2021; Pelto and Menounos, 2021; Van Tricht and others, 2021; Kneib and

others, 2022). Its state-of-the-art realisation is that of Kneib and others (2024), who reconstructed a high-resolution distributed SMB for Argentière Glacier (French Alps) over 2012–2021 from Pléiades elevation change and velocity, ground-penetrating-radar thickness and three ice-thickness models. Validation against the GLACIOCLIM stake network gave root-mean-square errors of 0.50–0.96 m w.e. yr^{−1} and $R^2 > 0.91$, resolving spatial patterns—including avalanche-fed accumulation hotspots—well beyond what stake networks capture.

Two difficulties beset this route. First, the flux divergence is the spatial derivative of the product of two independently-measured noisy fields, so it is dominated by small-scale noise that is suppressed by *post hoc* spatial filtering (Van Tricht and others, 2021; Bisset and others, 2020); yet smoothing a quantity that should satisfy mass conservation locally redistributes mass between grid cells and violates the very balance the method enforces. Second, and more fundamentally, the flux divergence is computed at a single observational snapshot and assumed representative of the whole observation period, whereas glacier geometry, velocity and flux all evolve under ongoing climate forcing (Jóhannesson and others, 1989; Zekollari and Huybrechts, 2015); treating the flux as stationary introduces a bias that grows with the length of the window—a limitation Kneib and others (2024) acknowledge explicitly. Per-glacier tuning of smoothing length scales, density assumptions and weighting schemes further hampers transfer to the regional scales most relevant for water-resource applications.

This study

Two features distinguish this formulation from flux-divergence inversions. First, the SMB is recovered by solving the forward mass-conservation equation rather than by differentiating observed fields: the flux divergence is recomputed at every time step from the evolving glacier state, so the inferred SMB is dynamically consistent over the whole window—rather than assuming a stationary flux—and mass is conserved exactly at every step, with no *post hoc* smoothing of a noisy divergence field. Second, we parametrise the SMB as a one-dimensional function of surface elevation, evaluated at every grid cell to yield a 2D field. A fully distributed SMB has as many unknowns as grid cells, whereas the geometric constraint supplies at most one scalar per cell—and none where the ice is thin or absent; collapsing the unknown to an elevation profile reduces the dimensionality from $\mathcal{O}(10^4)$ to $\mathcal{O}(10)$ and renders the otherwise severely ill-posed inversion well-constrained, without site-specific smoothing length scales or density assumptions. The price—horizontal variability within an elevation band, such as the avalanche-fed hotspots resolved by

Kneib and others (2024), is not recovered—is a deliberate trade-off of sub-elevation detail for a transferable, tuning-free inversion, to which we return in Section ??.

The accuracy of any geometry-based inversion is ultimately set by the input DEMs. Near-global elevation-change products (Hugonnet and others, 2021) make the method broadly applicable, but their smoothing limits the fidelity of the recovered SMB; dedicated multi-temporal DEMs deliver the highest accuracy, and the rapidly expanding archive of such data is what makes a regional-scale application realistic. We therefore validate the method with accurate two-epoch DEMs on three Alpine glaciers of contrasting size and data availability: Argentière (French Alps), where a dense stake network, repeated ground-penetrating-radar surveys and a rich Pléiades archive allow a direct benchmark against the state-of-the-art reconstruction of Kneib and others (2024); and Great Aletsch and Rhône (Swiss Alps), constrained by surface velocity alone, without any thickness survey, representing the data-sparse regime typical of most glaciers. Section details the two-step pipeline—data assimilation of the initial dynamical state, then SMB inference by automatic differentiation through the transient forward run; Section presents the inferred SMB profiles and their validation against in situ stake measurements; and Section ?? discusses the implications for regional-scale SMB inversion.

METHODS

Overview

We recover the surface mass balance (SMB) of a glacier with a two-step pipeline, implemented as a process module within the open-source Instructed Glacier Model (IGM; Jouvét, 2023; Jouvét and Cordonnier, 2023). The forward problem is governed by depth-integrated mass conservation,

$$\frac{\partial H}{\partial t} = b - \nabla \cdot \mathbf{q}, \quad \mathbf{q} = H \bar{\mathbf{u}}(H, z_s, c), \quad (1)$$

where H is the ice thickness, b the SMB, $z_s = z_b + H$ the surface elevation, c the basal sliding coefficient, and $\bar{\mathbf{u}}$ the depth-averaged horizontal velocity returned by IGM’s deep-learning ice-flow emulator (Jouvét and Cordonnier, 2023).

Equation (1) makes the central difficulty explicit: the geometry evolution depends jointly on the unknown SMB b and on the ice dynamics through the flux $\mathbf{q}$, which itself depends on H and c . Inferring b from an observed geometry change therefore requires the dynamical state to be fixed beforehand; otherwise

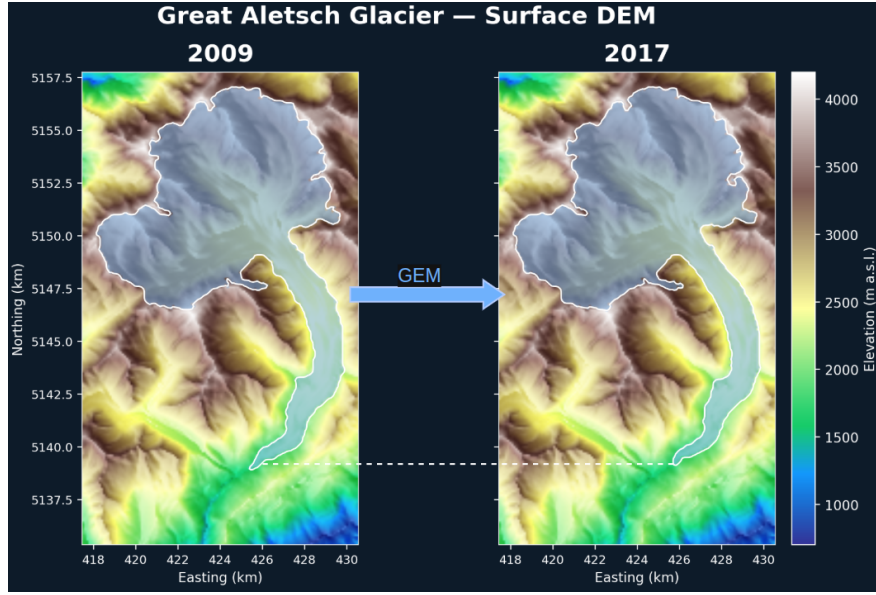


Fig. 1. Conceptual setup of the SMB-inference pipeline, illustrated for Great Aletsch Glacier. The panels show the observed surface elevation at the start (2009) and end (2017) of the inversion window. Prior to the inversion, data assimilation recovers the ice thickness—and hence the bedrock—at the 2009 epoch, constrained by the observed surface-velocity field (Millan and others, 2022) (Sect.). The transient glacier-evolution model (GEM) integrates the 2009 geometry forward to 2017 under a candidate elevation-dependent surface mass balance, and the inversion seeks the mass balance for which the modelled 2017 geometry reproduces the observed one (Sect.). The dashed line marks the observed 2017 terminus, highlighting the retreat of the tongue over the eight-year window. Colour denotes surface elevation.

an error in the dynamics would be aliased into the inferred SMB. Our pipeline reflects this dependence directly (Fig. 1): (i) we first calibrate the dynamical state by data assimilation against a surface-velocity snapshot near the start of the inversion window—inverting the ice-thickness field H , while the sliding and Arrhenius factor are fixed by an ensemble search—augmented on Argenti re, where extra observations allow the sliding coefficient to be inverted as a field (Sect.); (ii) holding the dynamics fixed, we then infer the SMB. Starting from the first surface DEM, we integrate (1) forward to the epoch of the second and seek the elevation-dependent SMB profile whose simulated geometry best matches the observed one (Sect.).

Ice-dynamics calibration via data assimilation

The calibration seeks the ice-thickness field that, when passed through the ice-flow emulator, best reproduces the observed surface velocities. We use IGM’s existing data-assimilation module (Jouvet, 2023) to invert for the thickness H by minimising the regularised misfit

$$\mathcal{J}_{\text{DA}}(H) = \alpha_u \|\mathbf{u}_s(H) - \mathbf{u}_s^{\text{obs}}\|^2 + \alpha_H \|H - H^{\text{obs}}\|_{\Omega_{\text{thk}}}^2 + \mathcal{R}(H), \quad (2)$$

where $\mathbf{u}_s(H)$ is the modelled surface velocity returned by the emulator, the second term penalises departure from point ice-thickness measurements over their survey footprint Ω_{thk} , and $\mathcal{R}$ is a Tikhonov-style smoothness regulariser on H . On Argenti re the thickness measurements are ground-penetrating-radar (GPR) soundings (Rabatel and others, 2018); for glaciers without such surveys (Great Aletsch, Rh ne) the second term is dropped and the calibration is governed by the surface-velocity misfit and the regulariser alone. The minimisation is solved by L-BFGS; we refer the reader to Jouv t (2023) for the full formulation, weighting strategy and convergence behaviour.

On the two velocity-only glaciers (Great Aletsch, Rh ne) the basal sliding coefficient c is *not* inverted pointwise. The emulated flow regime is instead set by two scalar physical parameters— c and the ice rate (Arrhenius) factor A —fixed beforehand by an ensemble search: candidate (c, A) pairs are each passed through the data assimilation, and we retain the pair that minimises the surface-velocity misfit, so the assimilation itself inverts the thickness H alone.

On Argenti re the GPR soundings supply an independent thickness constraint that makes a joint inversion identifiable, so we additionally relax c into a spatial field and invert it alongside H (the Arrhenius factor remains the scalar set by the ensemble search). This is also motivated physically: Kneib and others (2024) find that the spatial variability of the sliding coefficient helps explain the observed surface velocities on these glaciers, and the recovered field c_{DA} reproduces the expected structure (Sect.). In either case the smoothness weight in $\mathcal{R}$ is set at the corner of the resulting L-curve, and only once the calibrated state matches the observed velocity—and the GPR thickness, where available—to within observational uncertainty do we freeze the calibrated state and proceed to the SMB inversion.

SMB inference by automatic differentiation

Observation target. With the dynamics frozen, the inversion is driven by the ice-thickness field at the end of the window, T . From the calibrated thickness we fix the bedrock,

$$z_b(\mathbf{x}) = z_s^{\text{obs}}(\mathbf{x}, t_0) - H_{\text{DA}}(\mathbf{x}), \quad (3)$$

and, assuming z_b does not evolve over $[t_0, T]$ —justified on decadal timescales—combine it with the observed end-of-window surface DEM to obtain the target

$$H^{\text{obs}}(\mathbf{x}, T) = z_s^{\text{obs}}(\mathbf{x}, T) - z_b(\mathbf{x}). \quad (4)$$

The two surface DEMs $z_s^{\text{obs}}(\cdot, t_0)$ and $z_s^{\text{obs}}(\cdot, T)$ are the accurate, glacier-specific two-epoch elevation models described in Sect. ; their difference defines the geometry change the inversion must reproduce.

Forward model. With the dynamical state frozen at H_{DA} and the flow scalars (c, A) , the glacier evolves according to (1), which we integrate explicitly with a forward-Euler scheme,

$$H^{n+1} = \max(0, H^n + \Delta t^n [b^n - \nabla \cdot \mathbf{q}^n]), \quad (5)$$

where the non-negativity projection prevents spurious negative thicknesses near the margin. The step is limited by a Courant–Friedrichs–Lewy condition, $\Delta t^n = c_{\text{cfl}} \Delta x / \max_{\Omega} \|\bar{\mathbf{u}}^n\|$, with $c_{\text{cfl}} = 0.2$, and the flux divergence is evaluated with a centred finite-difference stencil on the staggered grid. Because $\bar{\mathbf{u}}$ is recomputed from the current (H, z_s) at every step, the flux divergence evolves with the geometry rather than being held at its initial-snapshot value—the property that removes the stationary-flux bias of geodetic flux-divergence inversions (Miles and others, 2021; Kneib and others, 2024).

SMB parametrisation. The unknown is a one-dimensional SMB profile $\beta(z)$, a function of elevation only, discretised at N uniformly spaced elevation bins by the values $\mathbf{m} = (m_1, \dots, m_N)$. At each time step it is mapped to the 2D source term in (1) by evaluating it at the current surface elevation, $b(\mathbf{x}, t) = \beta(z_s(\mathbf{x}, t))$; the interpolation is piecewise-linear and differentiable everywhere with respect to $\mathbf{m}$, so gradients propagate cleanly through every time step. As argued in Sect. , collapsing the unknown from $\mathcal{O}(10^4)$ pixels to $N = \mathcal{O}(10)$ bins reflects the first-order elevation control on alpine SMB and regularises the otherwise severely ill-posed inversion.

Inverse problem. We seek the profile $\mathbf{m}^*$ that minimises the misfit between the simulated end-of-window thickness $H^{\text{sim}}(\cdot, T; \mathbf{m})$ and the target $H^{\text{obs}}(\cdot, T)$ from (4):

$$\mathcal{J}(\mathbf{m}) = \sqrt{\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} (H^{\text{sim}} - H^{\text{obs}})^2} + \mathcal{R}(\mathbf{m}), \quad (6)$$

where the first term is the RMS geometry misfit and $\mathcal{R}(\mathbf{m})$ is a Tikhonov-style smoothness penalty on the squared discrete curvature of the profile; λ trades data fidelity against smoothness. The minimisation uses L-BFGS (?) with $M = 10$ correction pairs and a Hager–Zhang line search; the low dimensionality of $\mathbf{m}$ makes the implicit Hessian approximation accurate, and convergence is reached in $\mathcal{O}(10^2)$ function

evaluations. The gradient $\nabla_{\mathbf{m}}\mathcal{J}$ is obtained by reverse-mode automatic differentiation (Paszke and others, 2017) through the entire forward integration (1)–(5): every operation in the forward map—the emulator evaluation, the staggered-grid divergence, the explicit Euler update, the elevation-based interpolation, and the cost itself—is a differentiable TensorFlow operation, so the chain rule applies uniformly across the forward graph. The principal limitation is memory: a naïve AD trace stores the full computational graph, whose footprint scales as $\mathcal{O}(N_t \cdot N_\Omega)$ with N_t time steps and N_Ω grid points, exceeding the budget of a single GPU. We address this with gradient checkpointing (?): the integration is partitioned into $K = \sqrt{N_t}$ segments, only the segment-boundary states are stored during the forward pass, and the within-segment graphs are reconstructed during the backward pass, reducing the memory cost from $\mathcal{O}(N_t)$ to $\mathcal{O}(\sqrt{N_t})$ at the price of one additional forward evaluation per backward pass.

RESULTS

We evaluate the method on three Alpine glaciers spanning a wide range of size and data availability. Argenti re (French Alps) serves as a benchmark: among the most densely monitored glaciers in the world, it combines a long GLACIOCLIM stake record, repeated ground-penetrating-radar surveys and a high-resolution Pl iades archive, and it is the subject of the state-of-the-art distributed reconstruction of Kneib and others (2024), against which we compare directly. The Great Aletsch and Rh ne (Swiss Alps) represent the operational regime the method is ultimately intended for: neither has thickness soundings, so the data assimilation is constrained by surface velocity alone, and both are validated against raw stake measurements only.

The results are presented in three parts. We first report the Argenti re benchmark—the calibrated dynamical state, and the inferred SMB profile with its validation against the GLACIOCLIM network (Sects –). We then apply the identical pipeline, without re-tuning, to the two velocity-only glaciers (Sect.). Finally, we use these glaciers to quantify the sensitivity of the inferred SMB to the basal sliding coefficient—the dynamical parameter least constrained in the absence of thickness data (Sect.).

Calibrated ice-dynamics state

On Argenti re the data assimilation jointly retrieves the ice-thickness field H_{DA} and a basal-sliding field c_{DA} that reproduce both the observed surface velocity and the GPR thickness transects, converging within 10^2 L-BFGS iterations (Fig. 2). Modelled velocities match the observations to a root-mean-square residual of

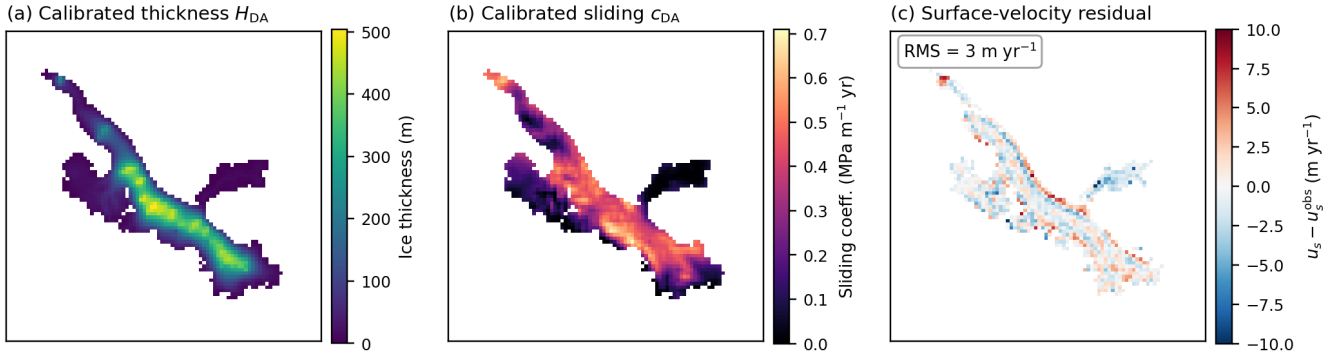


Fig. 2. Calibrated data-assimilation state on Argentière Glacier at $t_0 = 2012$. (a) Calibrated ice thickness H_{DA} and (b) calibrated basal-sliding field c_{DA} , the frozen dynamical state passed to the SMB inversion. (c) Surface-velocity residual $u_s - u_s^{obs}$, near zero across the glacier (overall $RMS = 3 \text{ m yr}^{-1}$); white cells lack a velocity observation.

3 m yr^{-1} , an order of magnitude below typical trunk velocities (20–80 m yr^{-1} , locally exceeding 150 m yr^{-1} on the Rognons confluence), and the calibrated thickness departs from the GPR transects by $RMS = 12 \text{ m}$ (bias -9 m), of the order of the GPR uncertainty itself (Rabatel and others, 2018). The sliding field shows the expected structure, enhanced along the central flowline and reduced near the lateral margins.

This calibrated state fixes the bedrock through Eqn. (3) and provides the frozen dynamical substrate for the transient forward run (Sect.). Integrated forward for one year under zero SMB it produces no spurious thickness drift, confirming that it is in mechanical equilibrium and that any subsequent geometry change is attributable to the prescribed SMB rather than to residual model imbalance.

Inferred SMB and validation

Holding (H_{DA}, c_{DA}) fixed, we run the transient forward simulation from 2012 to 2021 and invert for the elevation-resolved SMB profile $\beta(z)$ that minimises the geometry-misfit cost (Eqn. 6); the L-BFGS optimiser converges in $\mathcal{O}(50)$ function evaluations. The retrieved profile (Fig. 3) increases across the full altitudinal range of the glacier (1670–3610 m a.s.l.), from strongly negative values at the tongue ($-8.5 \text{ m w.e. yr}^{-1}$ at 1700 m) to a positive accumulation regime in the upper basin, and the simulated end-of-window thickness $H^{sim}(\cdot, T)$ matches the target $H^{obs}(\cdot, T)$ across the trunk to within the DEM differencing uncertainty (Fig. 4).

Validated against the GLACIOCLIM network— $\beta(z)$ sampled at the elevation of each stake on the main trunk and the Améthystes tributary, compared with the 2012–2021 mean stake measurements—the inferred profile reproduces the observed altitudinal pattern (Fig. 3) with a root-mean-square error of 0.454 m w.e. yr^{-1} (Table 1), lower than every previously published distributed reconstruction on this site

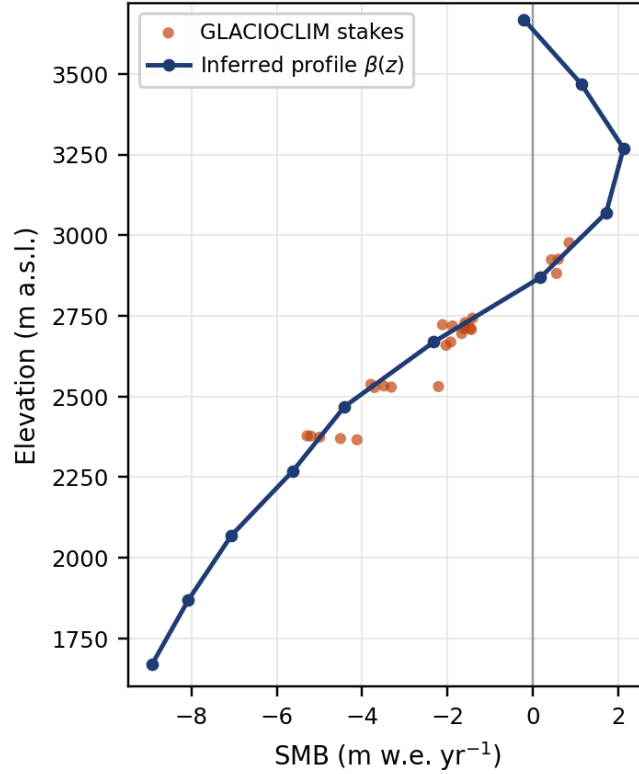


Fig. 3. Inferred elevation-resolved SMB profile $\beta(z)$ (blue line) for Argentière Glacier over 2012–2021, in water equivalent. Orange dots are the 2012–2021 mean GLACIOCLIM stake balances, plotted at the elevation of each stake. The profile is sampled at 11 elevation bins of 200 m width spanning 1670–3610 m a.s.l.

over this period: the F2019, SIA and IGM-distributed approaches of Kneib and others (2024) report 0.50, 0.67 and 0.85 m w.e. yr⁻¹ against the same stakes. Because the inversion variable is one-dimensional, horizontal heterogeneity within an elevation band—notably the avalanche-fed accumulation hotspots reported by Kneib and others (2024)—is not resolved, consistent with the spatially structured thickness residual in Fig. 4(c). This deliberate trade-off, which buys a tuning-free and transferable inversion, is taken up in Sect. ??.

A central feature of the method is that the flux divergence is recomputed at every time step from the evolving geometry, rather than held stationary at a single snapshot as in geodetic SMB inversions (Kneib and others, 2024; Miles and others, 2021). The bias this stationarity assumption would introduce is quantified directly against in situ stakes on the two velocity-only glaciers in Sect. (Fig. 5).

Transfer to glaciers without thickness data: Great Aletsch and Rhône

To test whether the method transfers to glaciers lacking thickness observations, we applied the identical two-step pipeline to Great Aletsch (2009–2017) and Rhône (2007–2016), changing only the input data:

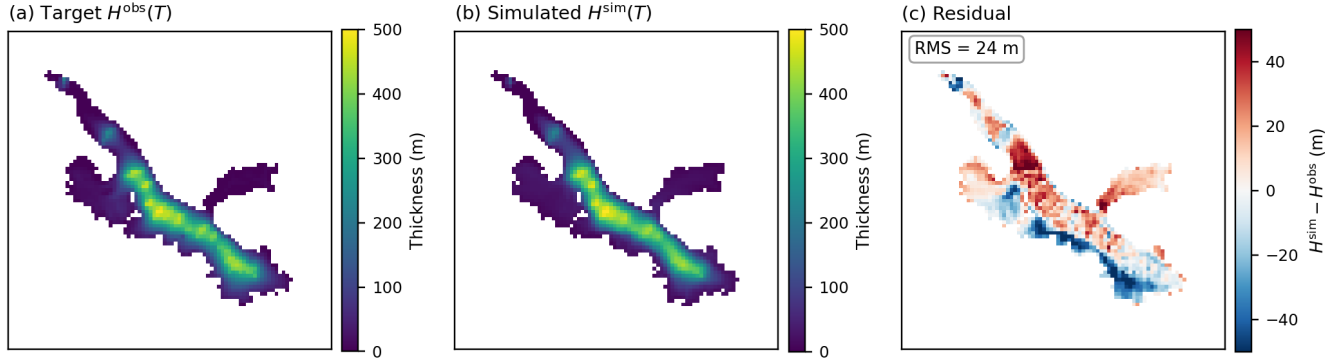


Fig. 4. End-of-window ice-thickness fit on Argentière Glacier ($T = 2021$), all fields masked to the glacier footprint. (a) Target thickness $H^{\text{obs}}(\cdot, T)$ from the observed surface DEM and the calibrated bedrock; (b) simulated thickness $H^{\text{sim}}(\cdot, T)$ from the forward run under the inferred SMB profile; (c) their residual $H^{\text{sim}} - H^{\text{obs}}$ (RMS 24 m over the glacier). The residual is small and largely unbiased; its spatially structured component reflects horizontal variability within elevation bands that the one-dimensional profile does not represent.

Table 1. Validation of inferred SMB against the GLACIOCLIM stake network on Argentière Glacier (main trunk + Améthystes tributary) over 2012–2021. The three baselines are the modelling approaches of Kneib and others (2024), evaluated against the same set of stakes.

Method	RMSE [m w.e. yr ⁻¹]
F2019 (Kneib and others, 2024)	0.50
SIA (Kneib and others, 2024)	0.67
IGM-distributed (Kneib and others, 2024)	0.85
Ours (1D AD inversion)	0.454

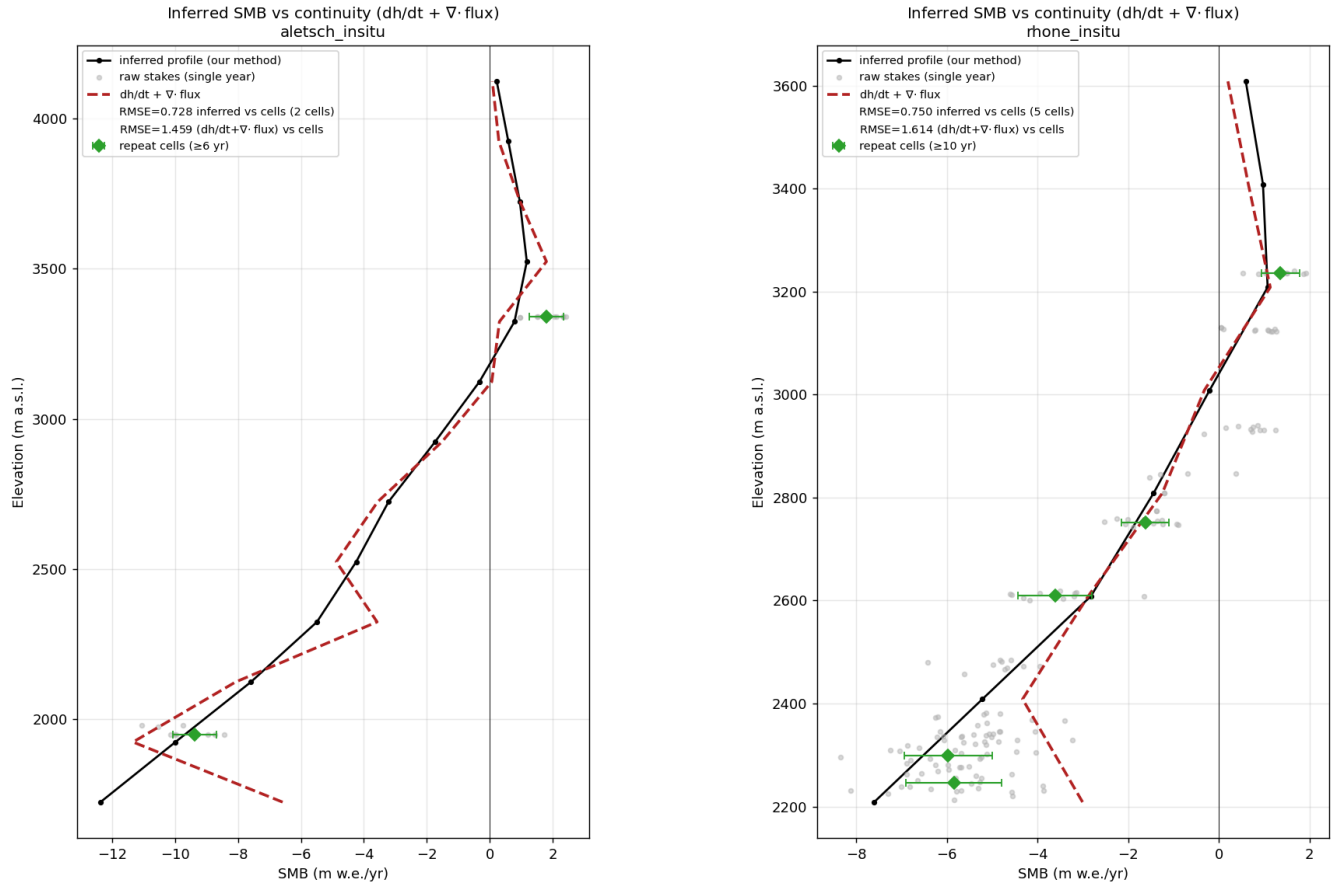


Fig. 5. Inferred SMB profile (black) for (a) Great Aletsch and (b) Rhône, compared against in situ stake measurements and against the continuity estimate $dh/dt + \nabla \cdot \mathbf{q}$ (dashed), obtained with the stationary data-assimilation flux divergence. Faint grey dots are raw single-year stake readings; green diamonds are multi-year repeat-cell means ($\pm$ interannual standard deviation), the validation targets. The inferred profile tracks the repeat-cell means across the whole elevation range, whereas the continuity estimate departs strongly in the ablation zone, where the stationary flux divergence is least representative.

no re-tuning of the regularisation, optimiser or parameterisation was performed. Because neither glacier has GPR coverage, the data assimilation is governed by the surface-velocity misfit alone (Eq. 2 with the thickness term omitted). The calibrated states reproduce the observed surface velocities to root-mean-square residuals of 9.9 and 6.3 m yr^{-1} on Aletsch and Rhône, respectively (Table 2), and the inversion converges to a comparable geometry misfit as on Argentière (Fig. ??).

The inferred profiles (Fig. 5) are smooth and monotonically increasing with elevation on both glaciers, from strongly negative balances at the tongue to positive values in the accumulation area, and reproduce the multi-year repeat-cell stake means—two cells on Aletsch and five on Rhône—with root-mean-square errors of 0.73 and $0.75 \text{ m w.e. yr}^{-1}$ (Table ??). These values are consistent with the Argentière benchmark despite the absence of any thickness constraint, confirming that the velocity-calibrated dynamical state is

Table 2. Sensitivity of the inferred SMB to the basal sliding coefficient c on the two velocity-only Swiss glaciers, with the rate factor A held fixed. *velsurf* is the data-assimilation surface-velocity misfit—the only selector available without stakes—while RMSE is the independent validation against multi-year repeat-cell stake means (n cells per glacier). Column minima per glacier in bold.

c (MPa)	<i>velsurf</i> (m yr ⁻¹)	$\bar{b}$ (m w.e. yr ⁻¹)	ELA (m a.s.l.)	RMSE (m w.e. yr ⁻¹)
<i>Great Aletsch</i> ($A = 182.1 \text{ MPa}^{-3} \text{ yr}^{-1}$; $n = 2$)				
0.098	15.59	-4.00	3009	2.83
0.154	9.94	-3.34	3127	0.90
0.199	10.14	-3.17	3183	0.73
0.269	11.36	-2.89	3198	0.86
<i>Rhône</i> ($A = 78.0 \text{ MPa}^{-3} \text{ yr}^{-1}$; $n = 5$)				
0.090	7.66	-1.62	2940	0.95
0.120	6.34	-1.88	2957	0.81
0.154	6.40	-2.05	2994	1.25
0.199	6.84	-1.72	2997	0.32
0.269	7.35	-1.72	2995	0.32?

sufficient to recover the elevation structure of the SMB.

Figure 5 also makes concrete the mass-conservation argument of Sect. . The dashed curve is the SMB that the continuity equation yields when the flux divergence is taken as stationary, exactly as in flux-divergence inversions. On both glaciers this estimate is markedly noisier than the inferred profile and diverges in the ablation zone—most severely on Rhône, where the dense, fast-thinning tongue drives the stationary divergence far from the stake means—raising the repeat-cell RMSE to 1.46 and 1.61 m w.e. yr⁻¹ (Table ??). Because our forward model integrates the mass-conservation law directly and recomputes the flux divergence from the evolving geometry at every step, the inferred profile requires no spatial smoothing and remains mass-consistent throughout, roughly halving the ablation-zone error relative to the stationary-flux estimate.

Sensitivity to the basal sliding coefficient

The basal sliding coefficient c is calibrated by data assimilation, but on a glacier without thickness or stake data its value rests on the surface-velocity misfit alone. To quantify how this choice propagates into the inferred SMB—and whether velocity data are sufficient to make it—we repeated the inversion on Aletsch

and Rhône over a range of c , holding the rate factor fixed at each glacier’s adopted value (Table 2). Both glaciers carry stake networks, so the stake RMSE provides an independent ground truth against the only selector available operationally, the data-assimilation velocity misfit (*velsurf*).

Three results emerge. First, *velsurf* is U-shaped in c and reliably rejects the corrupting regime: its worst value lies at the high-sliding extreme (lowest c), which on Aletsch is also where the SMB is most strongly biased—an over-deep ablation tongue with mean SMB -4.00 m w.e. yr $^{-1}$ and stake RMSE 2.83. Minimising the velocity misfit therefore keeps the inversion out of the regime that most corrupts the profile. Second, *velsurf* has a flat minimum and cannot pin a single optimum: the velocity-selected coefficient and the stake-optimal one are adjacent but not identical ($c = 0.154$ vs 0.199 on Aletsch, 0.120 vs 0.199 on Rhône), with the stake optimum lying within 2 % (Aletsch) and 8 % (Rhône) of the velocity minimum—inside the misfit noise. On Rhône the stake RMSE even varies non-monotonically across the basin (0.32 – 1.25 m w.e. yr $^{-1}$), confirming that velocity cannot resolve within-band differences. The data thus constrain c to a band, not a point. Third, this residual uncertainty maps almost entirely onto the ablation zone: across the flat velocity basin the accumulation-area profile is essentially unchanged, while the tongue SMB and hence the equilibrium-line altitude shift (3127 – 3198 m on Aletsch; ~ 40 m on Rhône), and the glacier-wide mean SMB ranges over -3.34 to -2.89 (Aletsch) and -2.05 to -1.72 m w.e. yr $^{-1}$ (Rhône) (Fig. 6).

Taken together, these results define a velocity-only procedure for stake-free glaciers: adopt the sliding coefficient at the velocity-misfit minimum, but propagate the two to three coefficients within ~ 10 – 15 % of that minimum into an SMB and equilibrium-line uncertainty band, tight in the accumulation area and widening at the tongue. The operational implications are taken up in Sect. ???. The rate factor was held at a temperature prior throughout; because it trades off with c in the velocity field, the band is conditional on that prior, which is the weaker of the two knobs.

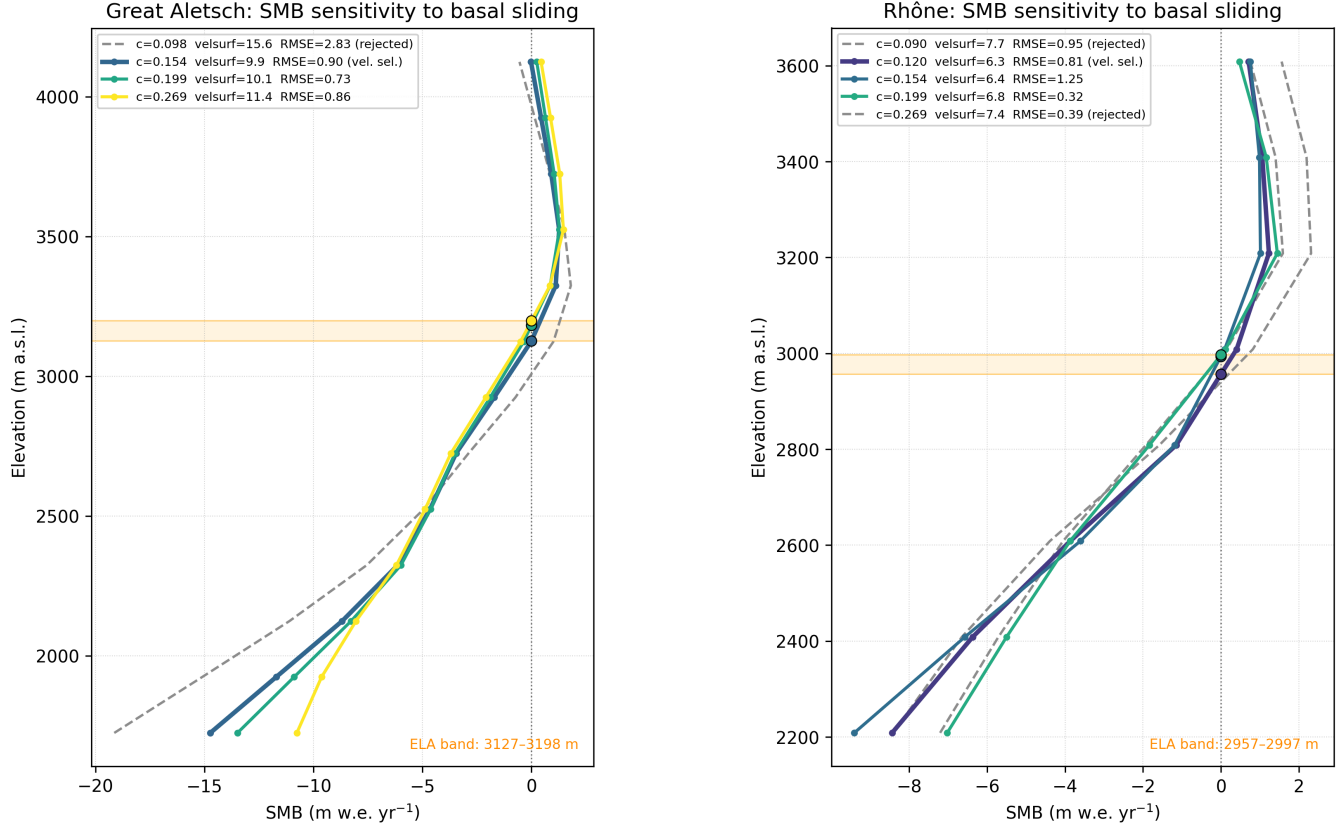


Fig. 6. Inferred SMB profiles for (a) Great Aletsch and (b) Rhône over the range of basal sliding coefficients within the velocity-misfit basin (Table 2). The profiles coincide in the accumulation area and diverge only at the ablation tongue, so the residual sliding uncertainty translates into an equilibrium-line altitude band (shaded) rather than a whole-profile bias.

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